The Impact of Large Language Models (LLMs) in the Health Sector

1. Introduction

- Background on LLMs
- Context & Focus
- Aim & Audience

2. Rationale & Significance

- Rationale for Topic Selection
- Need for the Review
- Research Gap

3. Methodology

- Scope & Search Strategy
- Inclusion/Exclusion Criteria:
- Data Extraction & Synthesis

4. Overview of LLM Development

- Historical Evolution
- Training Paradigms

5. Applications of LLMs in Healthcare

5.1 Clinical Decision Support

5.2 Medical Research & Data Analysis

5.3 Medical Education & Training

5.4 Patient Communication & Virtual Assistance

5.5 Emerging Multimodal & Domain-Specific Models

6. Benefits & Opportunities

- Efficiency & Accessibility

- Accuracy & Safety
- Innovation

7. Limitations & Challenges

- Hallucinations & Factuality
- Interpretability & Transparency
- Data Privacy & Ethical Considerations
- Evaluation & Regulation
- Resource & Sustainability

8. Thematic Comparison & Critical Analysis

9. Future Directions & Research Needs

- Standardised Benchmarks
- Multimodal & Interdisciplinary Models
- Policy & Governance
- Collaborative Design

10. Conclusion